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Researchers conducted a study to assess how sex category interpretation is related to implicit bias and cortical activity. For the first phase, researchers tested sex category interpretation using 12 highly realistic computer-generated "facial morphs" varying along a continuum of apparent sex category from extremely male (1) to extremely female (12). The sexually dimorphic content of the faces was manipulated, but all other aspects were unchanged. For example, eyes and noses remained constant, but brow lines and eyebrows were thickened for male faces. Participants viewed the 12 faces in random order, rating each face on a scale of 1 (extremely male) to 10 (extremely female). Despite the uniform increments in the sexually dimorphic content of the facial morphs, responses demonstrated dichotomous sex category classification, as expected.

For the second phase, which assessed the relationship between sex category interpretation and implicit bias, participants completed an implicit association test (IAT), in which they were shown the facial morphs with a pair of word attributes, one positive and one negative (eg, hard-working, lazy), and were asked to quickly pick the word they most associated with the face. Participants demonstrated shorter response times for facial morphs at the end points of the sex category continuum.

For the third phase, researchers assessed the relationship between face-sensitive regions in the occipitotemporal cortex, particularly the fusiform face area (FFA), and sex category interpretation. Participants were injected with a glucose analog containing a radioactive tracer an hour before brain activity was measured while participants viewed the facial morphs. Results suggest that FFA activity was strongest when the faces were at the end points of the sex category continuum, regardless of whether a face was perceived as male or female.

This study assesses how individuals interpret which of the following?

- ☐ A. Dyadic relationships [5%]
- ☐ B. Manifest functions [3%]
- ✓ ☒ C. Social identity [52%]
- ☐ D. Group identification [37%]

Explanation:

Personal identity vs. social identity			
	Comparison	Defined by	Self-concept
Personal identity	Intragroup: Different from other in-group members	One's self	Experience self as an individual
Social identity	Intergroup: Different from out-group members	Society	Experience self as a member of a group

Social identity describes how one's self-concept (ideas and beliefs about the self) is shaped by group membership. The major social identities are **sex/gender**, sexuality, race/ethnicity, socioeconomic status, age, family status (eg, married, parent), and occupation. Social identities define individuals **in relation to others** and allow for social groupings. This study assesses the interpretation of sex category, a social identity, by asking participants to categorize faces along a continuum of male and female.

Personal identity, on the other hand, describes how one's self-concept is shaped by more private or individual characteristics. For example, "Cubs fan" and "bookworm" are personal identities.

(Choice A) Dyads are the smallest sociological groups. Dyadic relationships characterize the interaction between the two people (eg, boss/employee, romantic partners). This study does not examine how two people interact.

(Choice B) Manifest functions are the intended consequences of a social structure. For example, the media (a social structure) is meant to disseminate information. This study does not address manifest functions.

(Choice D) Group identification refers to the extent to which an individual perceives himself or herself as a member of a larger collective. For example, identifying as a "pre-med student" associates an individual with a larger group of people who are studying for the MCAT and applying to medical school. This study assesses sex category interpretation but does not measure participants' *identification* with a specific sex category.

Educational objective:

Social identity refers to how individuals see themselves relative to others and is based on membership in socially defined groups. The major social identities are sex/gender, sexuality, race/ethnicity, socioeconomic status, age, family status, and occupation. Personal identities are subjectively defined, private or individual characteristics.

Time spent: 0 sec
 Subject:
 Behavioral Sciences

QID: 400693
 System:
 Identity and Social Interaction

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Which of the following statements is true? The principles of:

- ✓ ☒ A. social constructionism align with the distinction made in the first paragraph of the passage. [78%]
☐ B. symbolic interactionism align with the results of the third phase of this study. [6%]
☐ C. rational choice theory are the foundation for the implicit association test. [3%]
☐ D. structural functionalism explain why continuous variables are perceived categorically. [11%]

Correct

78% answered correctly

Explanation:

Major theoretical approaches to sociology			
Theory	Description	Perspective	Theorists
Functionalism	- Society = an organism - Each part of society works to maintain dynamic equilibrium (homeostasis)	Macro	Émile Durkheim, Talcott Parsons
Conflict	- Society = struggle for limited resources - Inequality based on social class	Macro	Karl Marx, Max Weber
Social constructionism	- Social actors define what is real - Knowledge about world based on interactions	Macro or micro	(Not critical for exam)
Symbolic interactionism	- Meaning & value attached to symbols - Individual interactions based on these symbols	Micro	Charles Cooley, George Herbert Mead
Rational choice/social exchange	Individual behaviors & interactions attempt to maximize personal gain & minimize personal cost	Micro	(Not critical for exam)
Feminist	Examines gender inequality in society	Macro or micro	(Not critical for exam)

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The first paragraph distinguishes between *sex*, *sex category*, and *gender*.

- **Sex** and **sex category** are defined by **biology** (sex organs, hormone levels) and **genetics** (presence or absence of the Y chromosome), even though not all bodies are strictly male or female (eg, intersex conditions).
- **Gender** is a **social construct** based on **social norms** and **roles** (expected behaviors for males and females).

Social constructionism is a sociological theory suggesting that "reality" is created through interactions, resulting in an **agreed-on shared meaning**. Objects (eg, money) and behaviors (eg, a handshake) have meaning only because individuals in society have agreed on that meaning. Money has value in society because everyone has agreed that it has value, thereby making money a **social construct**.

Social constructionism can focus on micro-level (individual) interactions or macro-level (large-scale) social structures. The individual *interpretation of sex category and gender* most aligns with the principles of social constructionism.

(Choice B) Symbolic interactionism is a micro-sociological approach emphasizing language and symbols and the subjective meanings attached to them. This theory focuses on the small daily interactions that take place between people and how these serve as the building blocks for society. Results from the third phase are specific to human perception and brain activity and do not deal with social interaction using symbols.

(Choice C) Rational choice theory is a micro-sociological approach suggesting that humans behave in ways that maximize personal gain and minimize personal loss. This theory proposes that behavior is rational and individuals act in their own self-interest, weighing the costs/benefits of each interaction. The IAT is designed to test *unconscious* assumptions and biases, not conscious, rational choices.

(Choice D) Structural functionalism is a macro-sociological approach emphasizing the ways that social structures work together to maintain equilibrium in society. This theory focuses on the large-scale forces (eg, education system, financial institutions) that shape society, not small-scale individual factors such as human perception.

Educational objective:

Social constructionism is a sociological theory suggesting that social constructs (ie, agreed-on meanings in society) determine small-scale (micro, individual) and large-scale (macro, structural) interactions in society. Sex category and gender are socially constructed.

Time spent:0 sec

Subject:
Behavioral Sciences

QID:400694

System:
Identity and Social Interaction

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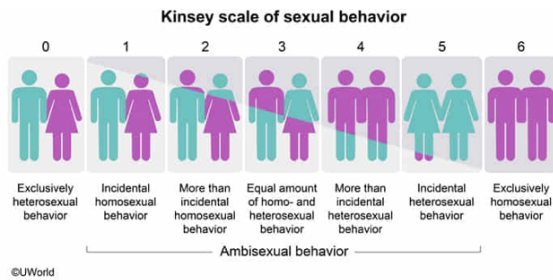
According to the first paragraph, the way people perceive gender is most analogous to the way people perceive which of the following?

- ✔ ☒ A. Sexual orientation [55%]
- ☐ B. Race [26%]
- ☐ C. Ethnicity [9%]
- ☐ D. Age [8%]

Correct

55% answered correctly

Explanation:



Sex (somewhat biologically defined), gender (socially defined), and sexual orientation are separate but related concepts. Sociologists define **sexual orientation** as a comparison between one's own sex category and the **sex category** to which one is **attracted** and with which one is **physically intimate**.

Like sex category and gender, sexual orientation is also a social construct that is often thought of as dichotomous, with homosexuality describing attraction to members of one's own sex category and heterosexuality describing attraction to members of the opposite sex category. However, there are a range of possible sexual behaviors, so theorists now believe that human sexuality, like gender, exists along a continuum, which can be visualized using the Kinsey Scale.

A major limitation of the Kinsey Scale is that it requires individuals to identify with a binary sex category. For example, for individuals who identify as intersex, it may not be possible to define sexual orientation as "same sex" or "opposite sex." Although there are a variety of additional labels for sexuality (eg, pansexual, sapiosexual, demisexual, asexual, queer) that do not fit along a heterosexual-homosexual continuum, the most common way of designating sexuality still relies on this method.

(Choices B and C) Race and ethnicity also exist along a continuum, but neither is considered dichotomous as there are more than two races and ethnicities.

(Choice D) Although age exists along a continuum, a person's age changes in annual increments. Age tends to be clustered into life-span stages (eg, toddlerhood, childhood, adulthood) and is not thought of as dichotomous or static, like gender.

Educational objective:

Sexual orientation describes people's sexual identity in relation to their own sex category and the sex category to which they are attracted. Sexual orientation exists along a continuum (eg, Kinsey Scale), even though it tends to be perceived as dichotomous (homosexual or heterosexual).

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Which neuroimaging technique was most likely used in phase 3?

- ☐ A. CT [4%]
- ☐ B. EEG [5%]
- ✓ ☒ C. PET [56%]
- ☐ D. fMRI [33%]

Explanation:

Neuroimaging method	Protocol	Measures
Electroencephalogram (EEG)	Electrodes placed on scalp & connected to an amplifier	Voltage fluctuations in the brain over time
Computerized tomography (CT)	Computer combines multiple x-rays taken at different angles	Detailed structure of internal organs & tissues at a single point in time
Functional magnetic resonance imaging (fMRI)	Scanner detects the differential properties of oxyhemoglobin & deoxyhemoglobin	Changes in blood oxygenation in the brain over time
Positron emission tomography (PET)	Scanner detects radioactive tracer attached to a glucose analog	Changes in glucose metabolism in the brain over time

Positron emission tomography (PET) is a **functional imaging technique** that monitors physiological activity by measuring labeled glucose metabolism. Active brain regions metabolize glucose more quickly, so a glucose analog with a radioactive tracer is used for PET neuroimaging. The metabolism of the analog results in the emission of positrons, which are detected by the PET scanner; this information is often combined with other imaging techniques to produce a three-dimensional image of the brain showing hypermetabolic and hypometabolic regions.

The final paragraph most clearly suggests the use of PET. It describes the injection of participants with a glucose analog containing a radioactive tracer an hour before brain activity measurement.

(Choice A) A computerized tomography (CT) scan uses x-rays to create cross-sectional images of the brain and is most useful for providing images of detailed structures. However, it is not useful for determining which part of the brain is active at any given time.

(Choice B) An electroencephalogram (EEG) measures electrical brain activity through the use of electrodes attached to the scalp. An EEG can provide some information about brain activity, but it cannot localize activity to a specific brain area, such as the fusiform face area (FFA).

(Choice D) Functional magnetic resonance imaging (fMRI) uses a powerful magnet to align the protons of water molecules in the body, creating nuclear magnetic resonance, which is measured by the scanner. This technique produces three-dimensional pictures of the brain and can indicate which brain areas are active at any given time. The use of a glucose analog containing a radioactive tracer is not necessary for fMRI.

Educational objective:

Positron emission tomography is a functional neuroimaging technique that measures

physiological activity in the brain. It uses a radioactive tracer attached to a glucose analog to measure which areas of the brain are metabolically active (using glucose more rapidly) during specific tasks.

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Which of the following displays the results of the first and second phases of the study, with the solid line representing participants' sex category ratings and the dashed line representing IAT response time?



Facial morph	Sex category rating (solid blue line)	Sexual selection (dashed orange line)
1	1.0	100
2	1.0	100
3	1.0	100
4	1.5	250
5	2.0	500
6	6.0	500
7	8.5	500
8	9.0	400
9	9.5	200
10	9.5	100
11	9.5	100
12	9.5	100

In the first phase, participants demonstrated a dichotomous sex category classification. Therefore, the solid line representing sex category ratings is expected to be sigmoidal, demonstrating an abrupt change in the middle of the range of facial morphs. Low sex category rating scores (male) should correlate with low facial morph numbers (male), and high sex category rating scores should correlate with high facial morph numbers, not the other way around (Choices C and D).

Graphs with two y-axes can be used to represent the relationship between one independent variable (x-axis) and two dependent variables (y-axes). Continuous variables that are perceived categorically appear to abruptly change from one extreme to the other, producing a sigmoidal graph.

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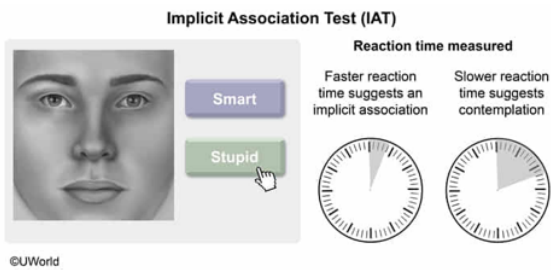
On the IAT, faster response times when pairing negative traits with less sex categorized faces most plausibly reflect:

- ☐ A. dissonance. [12%]
- ✓ ☒ B. social stigma. [62%]
- ☐ C. discrimination. [14%]
- ☐ D. functional fixedness. [10%]

Correct

63% answered correctly

Explanation:



A **stigma** is an attribute or behavior that is considered undesirable by most in society.

Social stigmatization is the **disapproval** of those with a **deviant characteristic** that does not conform to social expectations, such as an intersex appearance (combination of male and female characteristics). Social stigmas can result in prejudice (negative attitudes) or discrimination (unfair treatment) by others in society.

The Implicit Association Test (IAT) is a psychometric technique designed to measure attitudes that may be unconscious by requiring the participant to assign a word or concept to one of two opposite categories as quickly as possible. Faster response times suggest that an individual did not need to contemplate how to assign the word, so quickly assigning negative traits to less sex categorized faces suggests that the participant has an implicit belief that intersex appearance is bad. This implicit association could be plausibly interpreted as a social stigma associated with intersex (non-dichotomous) faces.

(Choice A) Cognitive dissonance is a state of mental discomfort caused by conflicting attitudes, beliefs, and/or behaviors. Slower (not faster) response times on the IAT are reflective of dissonance because they suggest that an individual is attempting to suppress an implicit bias, taking longer to respond.

(Choice C) Discrimination is the unfair treatment of an individual or group based on social characteristics (eg, race, sex category, gender, sexual orientation). The IAT is a measure of implicit attitudes or beliefs, which are reflective of *prejudice*, not discrimination.

(Choice D) Functional fixedness is a cognitive bias restricting the way one thinks of an object's uses to only typical or traditional uses. If someone needs a hammer but doesn't have one, functional fixedness would prevent the person from considering a paperweight or shoe to hammer in the nail. The IAT does not measure functional fixedness.

Educational objective:

Stigmas are characteristics that are defined as deviant and are therefore considered undesirable by most in society. Social stigmatization is the disapproval by others of individuals with these characteristics.

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